PROJECT ABSTRACT: LEAFLORE PERSONAL READING JOURNAL

Project Title: LeafLore: Personal Reading Journal (Gamified Tracker with ML & PWA)

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Core Trends: Machine Learning, Progressive Web Applications (PWA), Data Analytics

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Project Abstract

Maintaining consistent personal habits, particularly active reading, represents a major challenge due to the lack of engaging feedback loops in standard tracking applications. Traditional habit trackers rely on static progress bars and numerical tables, which often lead to user disengagement and habit abandonment. Furthermore, standard web-based trackers fail to operate under unstable network conditions, leading to data loss when users read during commutes.

To resolve these challenges, this project proposes LeafLore: Personal Reading Journal, a gamified reading habit tracker that integrates predictive analytics and offline-first mobile synchronization. LeafLore: Personal Reading Journal converts raw reading statistics into an interactive visual feedback system using an HTML5 Canvas-based Virtual Greenhouse Garden. As the user logs pages read, recursive graphic algorithms dynamically grow sprouts, trees, and foliage in real-time, accompanied by unlockable themes and achievements to drive user motivation.

At its core, LeafLore features an intelligent Completion Forecast Engine. When a user has a mature reading history, the system trains a Logistic Regression Machine Learning Model (via Scikit-Learn) using features such as reading velocity, focus session ratios, and note-taking frequencies to predict the likelihood of completing a book on time. For new users with sparse histories (the cold-start problem), the system dynamically transitions to an analytical heuristic pacing algorithm.

To ensure uninterrupted accessibility, LeafLore is engineered as a Progressive Web App (PWA). Using custom service worker caching strategies and browser-level IndexedDB transactional databases, users can log pages and save focused reading sessions offline. Once network connectivity is restored, the client-side queue automatically synchronizes with the server database.

LeafLore demonstrates a modern, deploy-ready solution combining machine learning classifications, data aggregation, and mobile-first offline architectures to foster sustainable habit formation.

Key Technologies & Frameworks

Layer / Component	Technologies & Libraries utilized
Backend Web Engine	Python, Flask framework, WSGI configuration (Gunicorn)
Database & ORM	SQLAlchemy ORM, Flask-Migrate (Alembic), SQLite & PostgreSQL
Machine Learning	Scikit-Learn (Logistic Regression model), Pandas & NumPy
Frontend Visuals	HTML5 Canvas API, CSS3 3D transforms, CSS design system
Offline PWA Caching	Service Worker caching (sw.js), manifest.json, IndexedDB API